



Microsoft Fabric

Microsoft Fabric Data Warehouse Analytics

Proof-of-Concept (POC)



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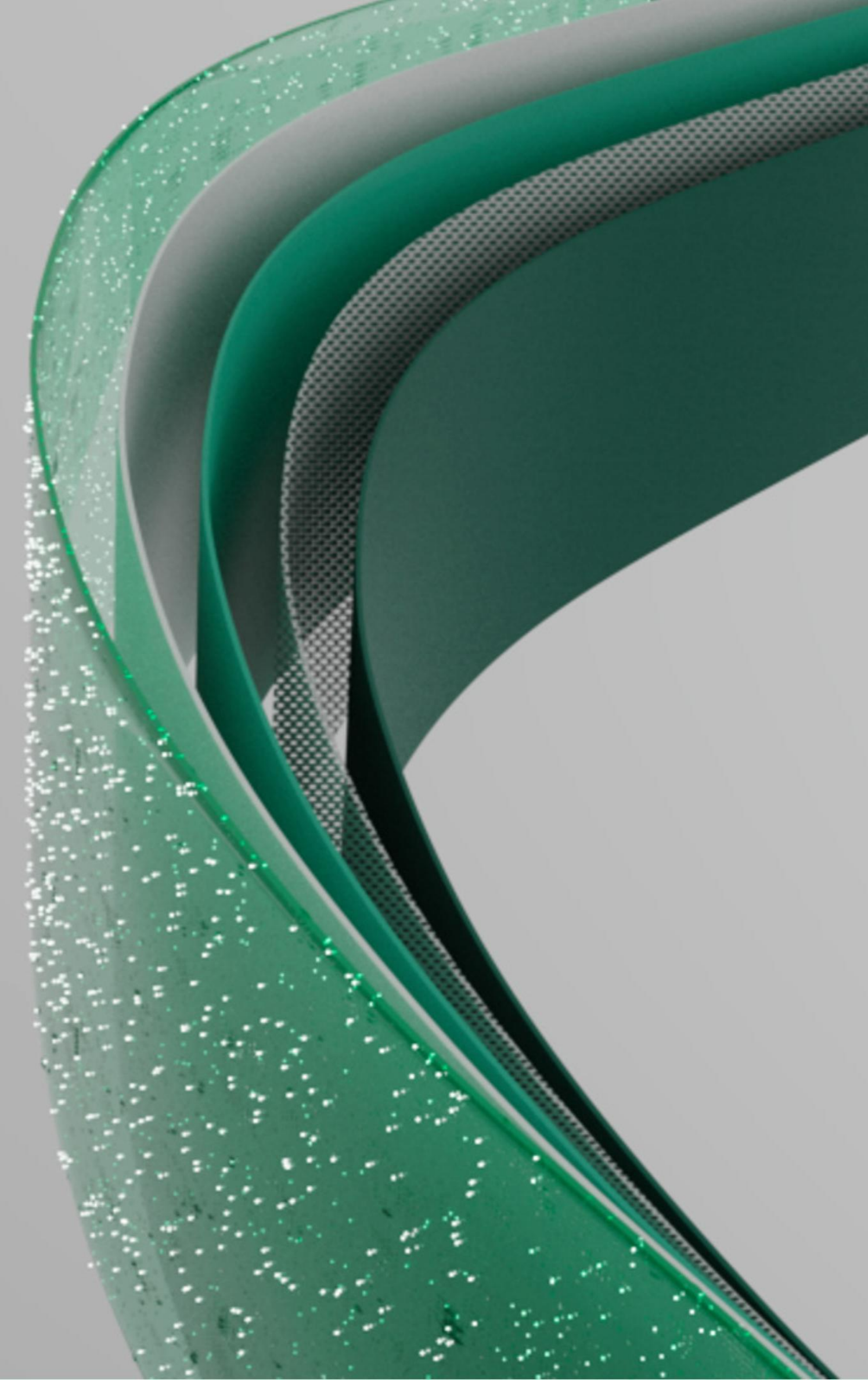
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PHASE 1: ENVISION

The art of the possible

What are your customers, employees, partners and suppliers' unmet needs and how can you uniquely solve those unmet needs through unified data, comprehensive insights and faster decision making? Let's have a closer look at what Microsoft Fabric makes possible for the following industries.

Healthcare

Financial Services

Public Sector

Retail

Sustainability/Energy



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Experiment & Learn



Build & Scale



Financial Services

The financial services industry (FSI) is heavily regulated and governed, which makes being backed by the security of a Microsoft product that much more important. With Microsoft Fabric, FSI professionals can access customer portfolio and market data with open and governed access controls. They can also strengthen risk prevention by leveraging scalable compute and analytics power modeling.

In the banking industry, organizations will be able to drive increased customer lifetime value and loyalty by providing a holistic view of the customer's current

financial state, allowing them to upsell or cross-sell products and services that best fit those individual customer needs.

In the insurance industry, organizations will be able to gain insights about customers by connecting internal and external data to provide custom solutions to the customer.

In capital markets, organizations can initiate more personalized client interactions by gaining more insights into a client's profitability and predictive analytics, enabling them to sell the right product at the right time to close more deals.

Data challenges

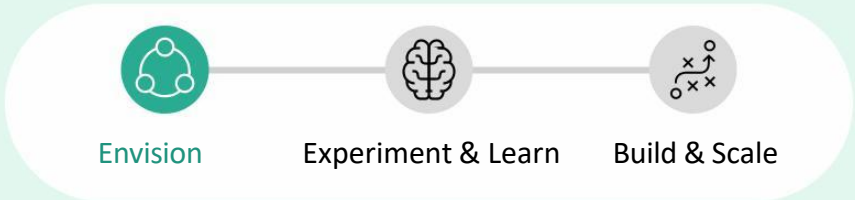
- Abundance of data and technology siloes
- Securely collecting, storing, and sharing data in a highly regulated industry
- Slow time to informed decision-making and risk responses

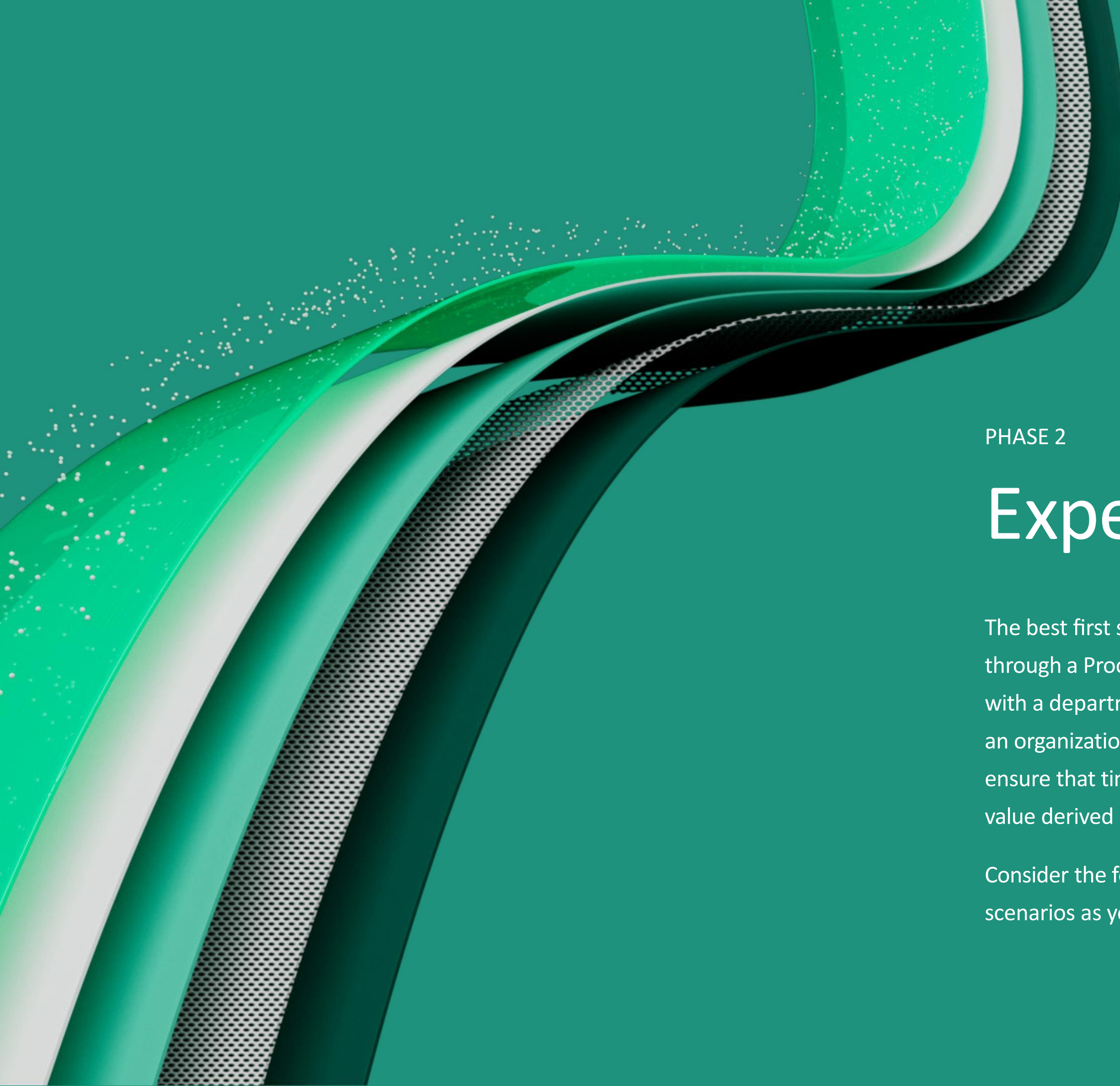
How Microsoft Fabric can help

- Collect data in Data Warehouse to provide one unified source of truth
- Access all portfolio, reference, market, and risk data with open and governed access controls
- Analyze all different types of historical data, using built-in security and governance policies

Key outcomes

- Enable decision-making backed by data to thrive in a competitive environment
- Gain a complete, 360-degree view of customers through secure data sharing
- Drive accurate insights for Risk Management and Compliance Reporting





PHASE 2

Experiment & Learn

The best first step is to experiment and learn through a Proof of Concept (POC). Starting with a departmental scenario rather than an organization-wide scenario can help you ensure that time spent is minimized and the value derived is maximized.

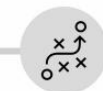
Consider the following departmental scenarios as you embark on your POC.



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Operations



Undergo production optimization

- Collect and analyze data from different manufacturing centers to identify production bottlenecks before they make an impact on the business.
- Develop predictive models to help them understand when to scale production up and down based on market trends. Increase efficiency by identifying production bottlenecks faster using a heads-up dashboard and custom diagrams.
- Use visualizations to easily understand data from manufacturers or production lines in one dashboard to aid in planning.
- Understand anomalies in historical manufacturing data using drill-down capabilities, which better informs required corrective action.



Enhance inventory planning and performance

- Unify sales data along with production data to identify which products to prioritize based on trends in the market and address stock levels before inventory shortages hit.
- Analyze which products to prioritize by easily spotting trends, such as regional spikes in demand or increased inventory cycle times for certain items.
- Assess and adjust levels of safety stock by combining recent trend and historical data from cloud, on-premises, or hybrid data sources.
- Receive notifications of major changes in available inventory across different geographies by configuring mobile alerts to help avoid inventory shortages.



Improve sales and operations planning

- Merge sales, marketing, and manufacturing data from different clouds and systems of record to draft new product development or production plans more easily.
- Improve sales and production forecasting quality by analyzing metrics like past total sales, production output and shipment times together using natural language querying or other advanced techniques to move from hindsight to insight.
- Get better visibility into demand and supply projections across departments from anywhere with all data stored in the same logical data lake.



Streamline fulfillment and distribution

- Make it easier to identify over- and under-used resources using production pick rates to predict whether to flex up or down and assess distribution center efficiency.
- Notify your customers sooner when there is a shipping delay by leveraging alerts from dashboards that can inform you about a shipping gap.



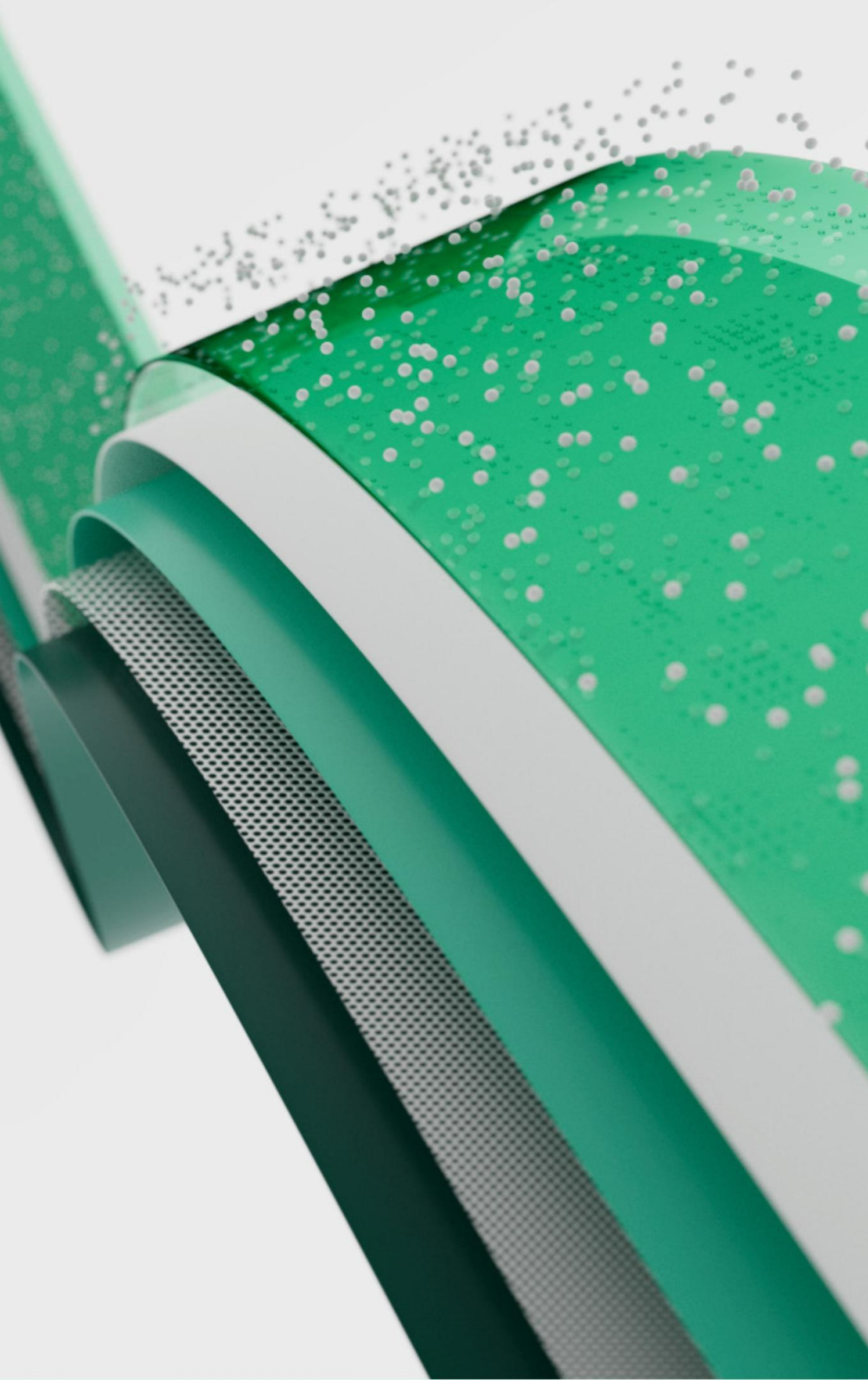
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Identify technical experts and business users for the POC

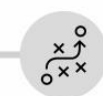
Identify availability of technical experts and business users to support you during the POC execution. Ideally, the technical decision makers in your organization, either an IT leader and/or a data leader, will start by creating clear requirements for the proof of concept. Next, your data practitioners such as architects, engineers, data scientists and analysts will execute the POC. And lastly business decision makers or lines of business (LOB) leaders will play a large role in evaluating the results.



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WORKSHEET

Document clear goals for your POC and the criteria that will define its success

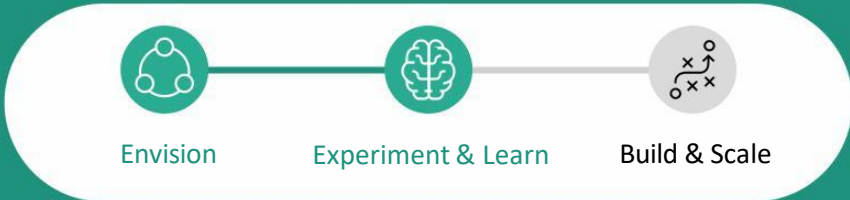
Keep in mind that a POC should be a short and focused effort to quickly prove a limited set of concepts and capabilities. These concepts and capabilities should be representative of the overall workload. Start by creating goals for your POC.

Use your goals to identify specific tests and to provide the outputs you identified. It’s important to make sure that you have at least one test to support each goal and expected output.

Goal	Output	Test 1	Test 2

A few suggested goals

- Reduced number of products used from X to Y
- Time saved from X to Y
- Improve development time from X to Y
- Expand analytical potential between X and Y through the addition of the new data
- Expand user base from X to Y
- Satisfy user base demonstrated by X score on user surveys
- Improve dashboard performance from X to Y
- Decrease time to insight from X to Y
- Business value added. This should be specific to the scenario you choose
- Performance benchmarking
- Low code/no code data pipelining



Using common analytics scenarios and proposed activities to guide your Proof of Concept

Microsoft Fabric supports four common end-to-end analytics scenarios:

1. Lakehouse
2. Data Warehouse
3. Data Science
4. Real Time Analytics

Please consider these scenarios as a guide to support your proof of concept. However, keep in mind that while these are the most common scenarios supported by Microsoft Fabric, there are many others that technical and business end-users can navigate through.



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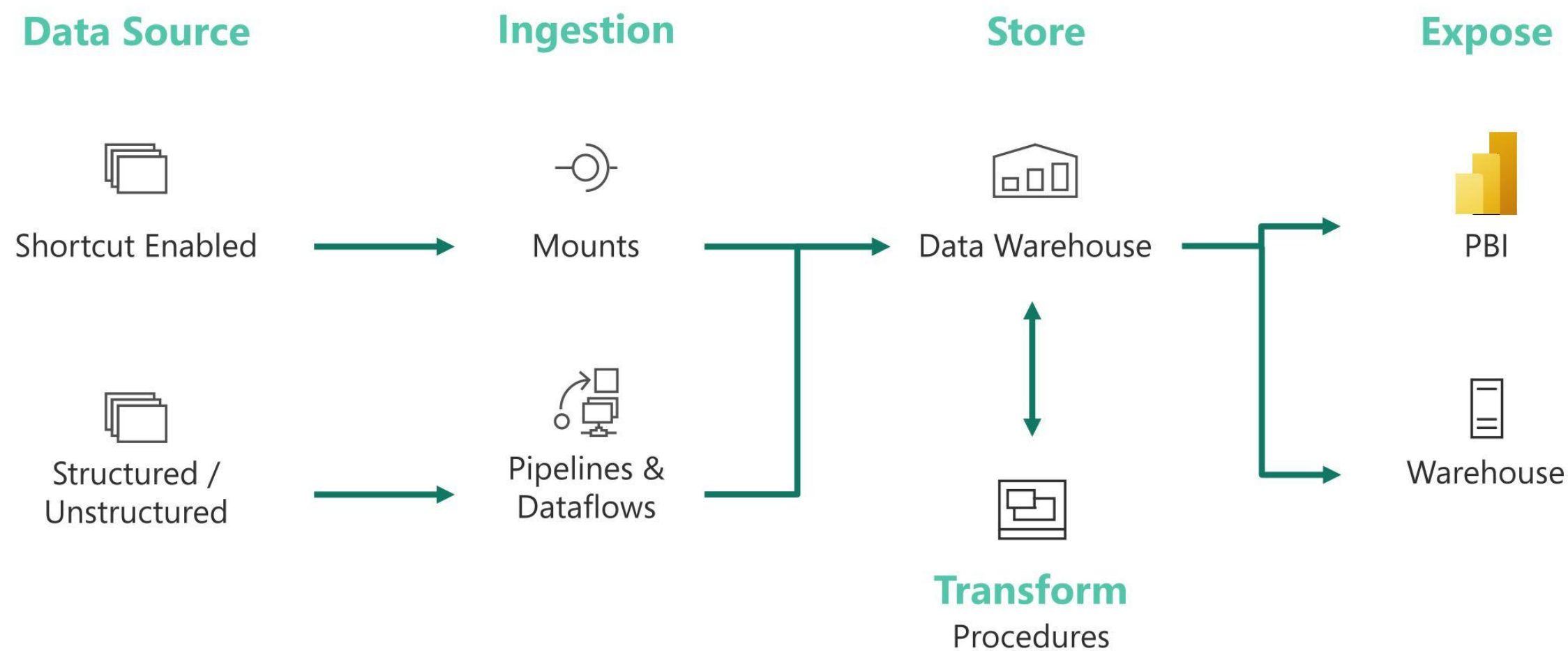
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Data Warehouse

The Data Warehouse analytics scenario takes existing sources that are mounted, while pipelines and dataflows can bring in all other data that is needed.

IT teams can then define and store procedures to transform the data, which is stored as Parquet/Delta Lake files in Data Warehouse.

From there, business users can analyze and visualize data with Power BI, again using the Direct Lake mode or SQL endpoints.



Build and implement an end-to-end data warehouse for your organization:

1. Enable Microsoft Fabric in your tenant
2. Create a Fabric workspace
3. Quickly create a data warehouse
4. Ingest data from source to the data warehouse dimensional model
5. Transform the data to create aggregated datasets using T-SQL
6. Perform orchestration, data ingestion, and data transformation with pipelines
7. Query the data warehouse using T-SQL and a visual query editor
8. Create Power BI report using Direct Lake mode to analyze the data in place
9. Cleanup resources by deleting the workspace and other items



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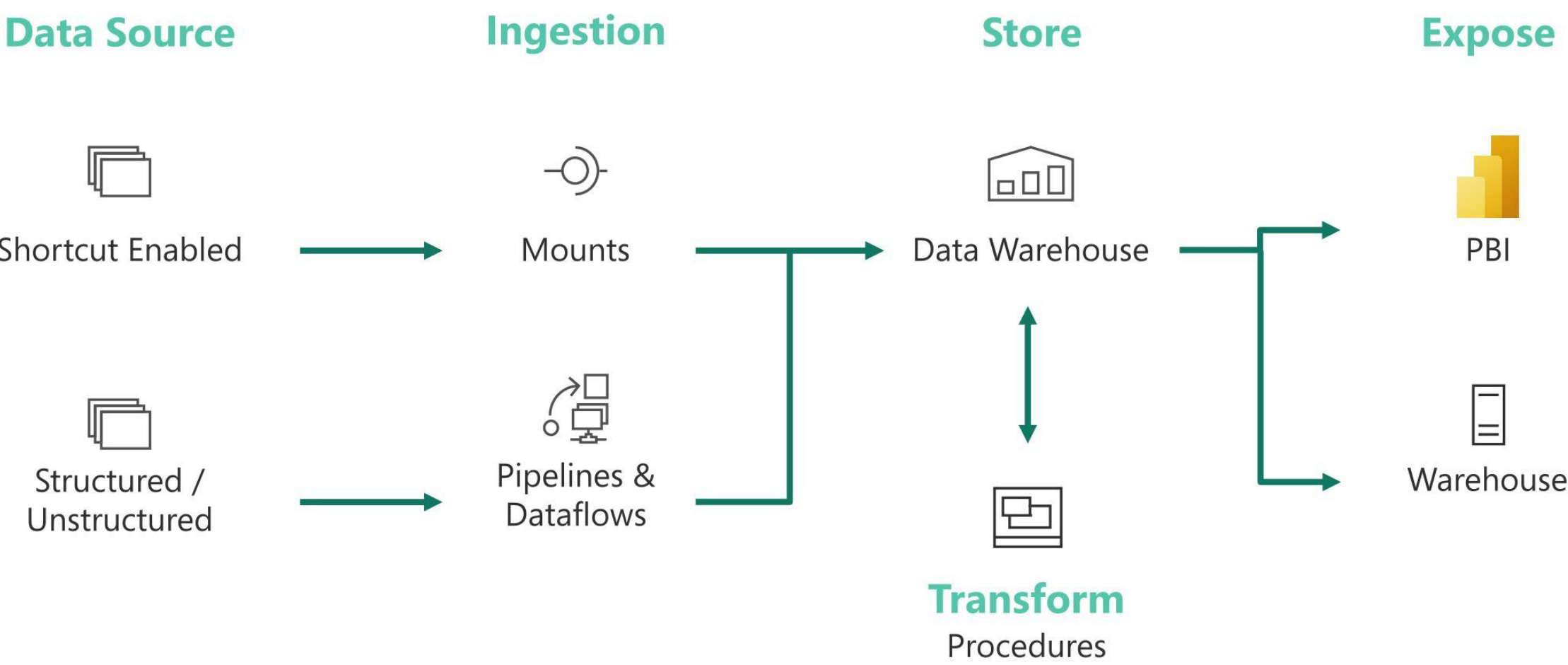
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Data Warehouse

The data warehouse end-to-end architecture:



Data Sources – Microsoft Fabric makes it easy and quick to connect to Azure Data Services, other cloud platforms, and on premises data sources to ingest data from.

Ingestion – With 200+ native connectors as part of the Microsoft Fabric pipeline and drag and drop data transformation with dataflow, you can quickly build insights for your organization. Shortcut is a new feature in Microsoft Fabric that provides a way to connect to existing data without having to copy or move it.

Transform and Store – Microsoft Fabric standardizes on Delta Lake format, that means all the engines of Microsoft Fabric can read and work on the same dataset stored in OneLake – no need for data duplicity. This storage allows you to build a data warehouse or data mesh based on your organizational need. For transformation, you can choose either low-code or no code experience with pipelines/dataflows or use T-SQL for a code first experience.

Consume – Data from your lakehouses can be consumed by Power BI, an industry leading business intelligence tool for reporting and visualization. Each data warehouse comes with a built-in TDS/SQL endpoint for easily connecting to and querying data from other reporting tools, when needed. When a data warehouse is created a secondary item, called a default dataset, will be automatically generated at the same time with the same name of the data warehouse to start visualizing data with just a couple of mouse clicks.



PHASE 3

Build & Scale

Delivering accelerated data value is not an overnight process. It involves significant organizational alignment. It necessitates building a solid and reliable data foundation. And lastly it requires consistently delivering business value and enabling decision making to build trust through incremental success over time.



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Building a foundation of alignment

We recommend the following steps to ensure the foundation for your journey is solid and you can help your entire organization embrace the culture of insight and innovation.

1. Get aligned with your organization's leaders and data stewards

- Build confidence with department leaders
- Collaborate on your business value roadmap
- Develop your infrastructure roadmap

2. Map your current and future data estate

- Document the details of your current data estate
- Ideate on planned transformation and outcomes

3. Establish a network of transformation champions

- Assemble a team of cross-functional subject matter experts



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Aligning with your organization's leaders and data stewards



4X

Alignment with your organization's department leaders is crucial to a successful, iterative and value-driven data estate transformation. You'll likely need to start by helping these important stakeholders see the broader vision of what is possible and the economic impact associated with your aspirations.

According to the Harvard Business Review, data shows that "companies that embrace a data-driven culture experience a 4x improvement in revenue performance and better customer satisfaction". The impact of a data culture on an organization is enormous.

[Source: How to lead a data-driven digital transformation by Harvard Business Review](#)

Stakeholders crucial to data estate transformation

- Chief Financial Officer and Other Finance Leaders
- Chief Sales Officer and Other Sales Leaders
- Chief Marketing Officer and Other Marketing Leaders
- Human Resources Leaders
- Chief Operations Officer and Other Operations Leaders (including Supply Chain, Ecommerce, Manufacturing, and more)
- Customer Service Leaders
- Research and Development Leaders
- Corporate Strategy and Planning Leaders



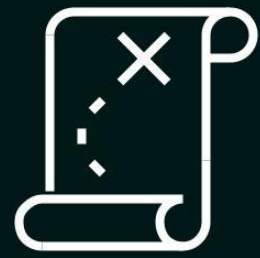
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Conversations and questions to drive your business value roadmap

We recommend running ideation sessions with department leaders to identify the initiatives that can drive the most business value once your data estate is unified. Consider an initiative that can drive some initial momentum and can be iterated on.

1. How can we enable innovation with data and analytics for your department?
2. What innovation can we enable?
3. How can I make your line of business more efficient?
4. What immediate initiatives can we collaborate on and how will we measure the impact?
5. What business value will we deliver over time? Cost reduction? Revenue generation?
6. How can I better enable your line of business to use data in decision-making?
7. What security and governance concerns should we be aware of?
8. Is there third-party or historical data that could enrich our insights?



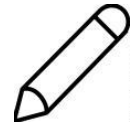
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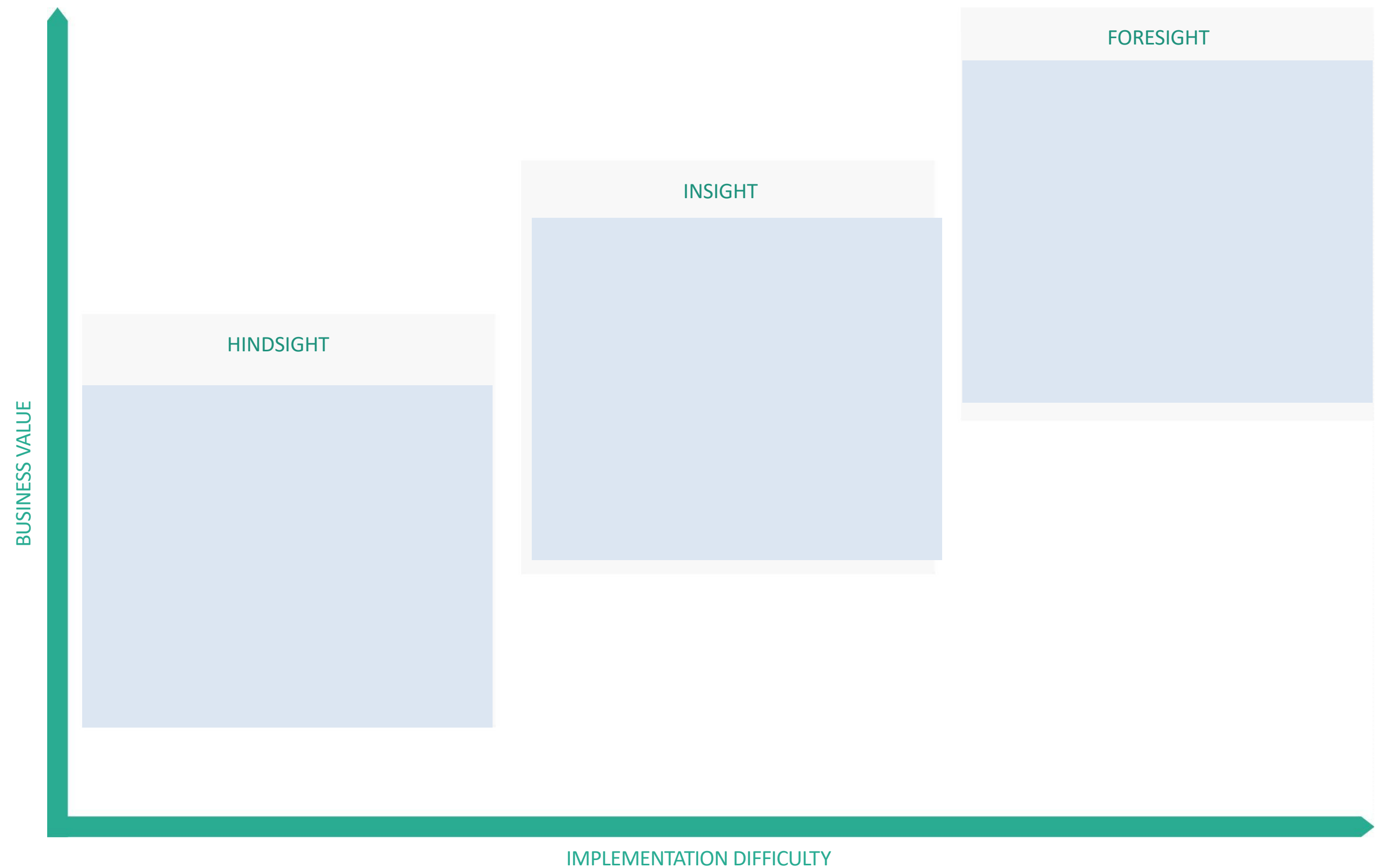
Create a list of your initiatives to drive business value. Categorize each initiative by the type of analytics it will support in the future.

Hindsight: is both descriptive and diagnostic, answering what happened and why it happened

Insight: is predictive, to consider what might happen

Foresight: is prescriptive, to suggest a course of action

This will be the start of your backlog and will help you prioritize based on the impact it has the potential to make. While you will ultimately settle on one or two ideas to begin with, once you have started and begun seeing success, this backlog of ideas will form a roadmap of what to do next.



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Conversations and considerations to drive your infrastructure roadmap

Alignment between the Chief Data Officer, Chief Technology Officer, and Chief Information Officer is imperative in the transformation of your data estate. Ensuring partnership through clear functional alignment as well as decision-marketing authority during these initiatives will pave the way for a smooth journey.

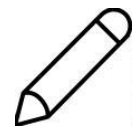
Key Considerations and Discussion Points for Alignment

- Budget Ownership
- Decision Making R&R
- Governance
- Compliance
- Quality Assurance
- Operating Models
- Processes
- Monetization

Data & Cloud
Engineering,
ETL, Data Lakes &
Warehouses

Insights

Data Sciences, AI/
ML



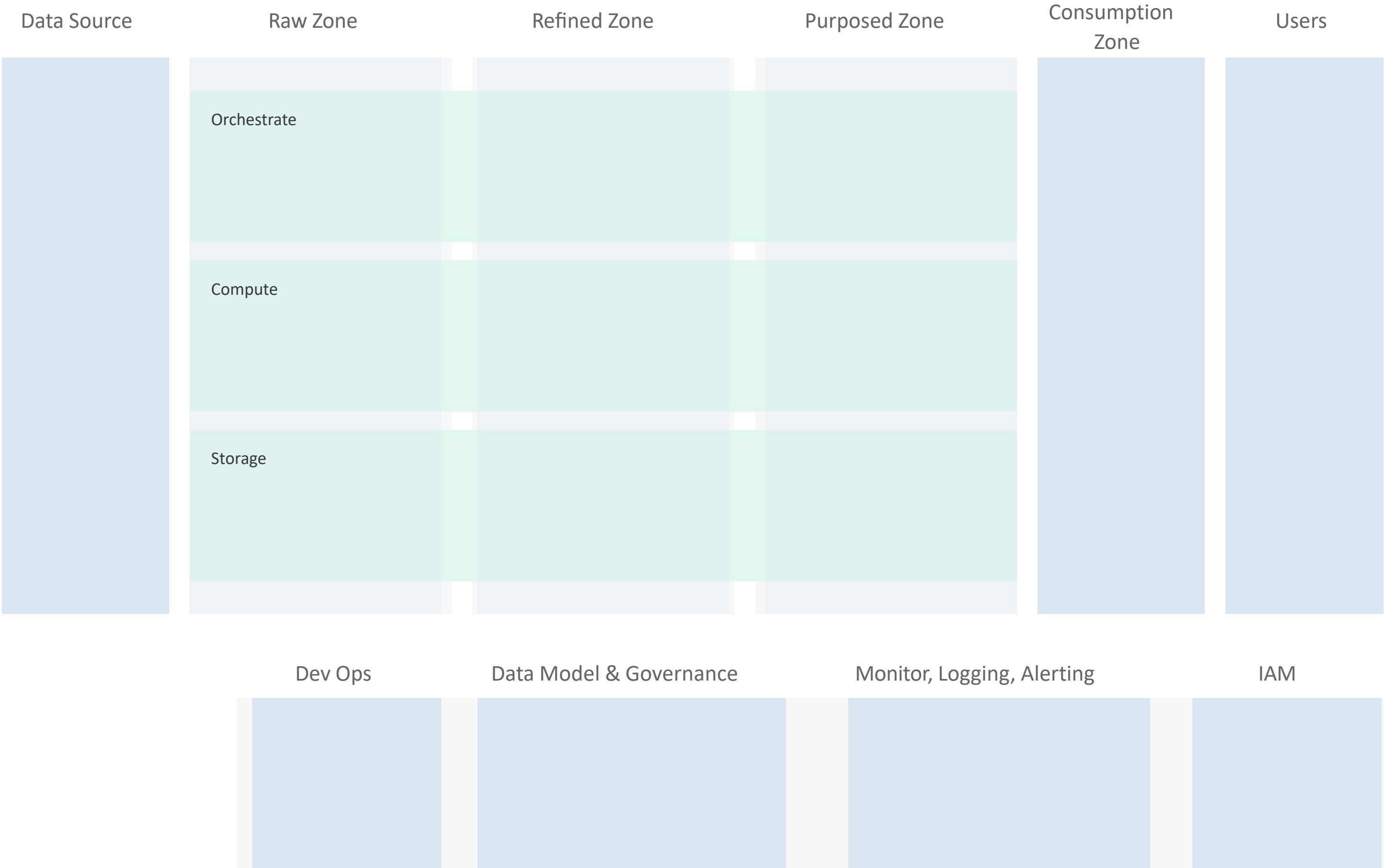
WORKSHEET

Mapping your current data and analytics estate

With your infrastructure team, begin documenting your current data estate. Be sure to capture which data sources, technologies, ELT/ETL scenarios, data pipelines, and appliances exist. Also, document how data is consumed by users.

Document the details of your data estate.

Which data sources, technologies, ELT/ETL scenarios exist? How is data consumed by users?



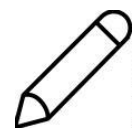
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Ideate on planned transformation and outcomes

Begin to document the details of your data estate transformation. Use the inputs from your business value roadmap to ensure you’ve prioritized these initiatives in your phased infrastructure evolution.

Technology	In Scope/Out of Scope	Planned Transformation	Expected Outcomes



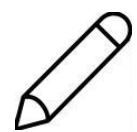
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Establishing a network of transformation champions

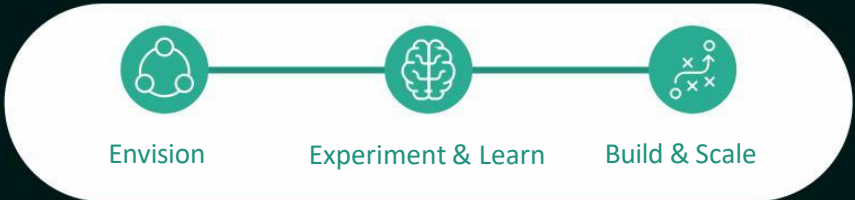
Begin assembling your team of champions for your data estate transformation. Select people who will work well together, but also come from different roles and competencies in the organization and with various skills and experiences. You'll want to find a balance between functional expertise.

Team Member	Role	Functional Skillset



We recommend the following roles as part of your innovation team.

- Department leaders (VPs / Directors): who can bring significant domain expertise
- IT leaders: Infrastructure admin and developers
- Data practitioners: Data scientists, data engineers, business analysts, data architects who will bring specialized knowledge of a process, function, or task
- Data stewards: Data professionals with knowledge of security, compliance, governance, and access management
- Business users and data analysts: Power BI power users



Training your network of champions

To help your team get more familiar with Microsoft Fabric, we've put together learning modules



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You are on your way to transformative data value

Congratulations on following the journey to data estate transformation. You are well on your way to leading your organization to accelerated data value and innovation. As you continue your path, you may find these additional resources helpful.

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